Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

Helene Bei Thomsen

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HEALTH

Supervisors and Assessment Committee

PhD student:

Helene Bei Thomsen
Department of Public Health, Aarhus University, Denmark

Supervisors:

Associate Professor Adam Hulman, PhD (main supervisor)
Department of Public Health, Aarhus University, Denmark
Steno Diabetes Center Aarhus, Aarhus university Hospital Denmark

Anders Aasted Isaksen, MD, PhD
Steno Diabetes Center Aarhus, Denmark

Guy Fagherazzi, PhD
Director of the Department of Precision Health
Head of the Deep Digital Phenotyping Research Unit
Luxembourg Institute of Health, Luxembourg

Signe Toft Andersen, MD, PhD
Steno Diabetes Center Aarhus, Denmark
Gødstrup Hospital, Denmark

Assessment committee:

Associate Professor Simon Tilma Vistisen (chairman)
Department of Clinical Medicine, Aarhus University, Denmark

Professor Cristina Soguero-Ruíz
Department of Signal Theory and Communications, King Juan Carlos University, Spain

Associate Professor Simon Lebech Cichosz
Department of Health Science and Technology, The Faculty of Medicine, Medical Informatics and Image Analysis, Aalborg University, Denmark

Disclosures and Notes

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SDCA Steno Diabetes Center Aarhus

Notes on website version

This dissertation is available for viewing online at HeleneThomsen.github.io/dissertation/ or by scanning the QR code in the lower right-hand corner of the title page.

Acknowledgements

Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. JMIR Diabetes, 11(1), e89374. https://doi.org/10.2196/89374

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at JMIR AI, (7.30.2026), pending for preprint at JMIR preprints)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. PLOS Digital Health, 4(6), e0000918. https://doi.org/10.1371/journal.pdig.0000918

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Abbreviations

AGP: ambulatory glucose profile

AID: Automated insulin delivery

AUROC: Area Under the Receiver Operating Characteristic Curve

CGM: continuous glucose monitoring

CV: coefficient of variation

CVD: cardiovascular disease

eGFR: Estimated Glomerular Filtration Rate

GMI: glucose management indicator

HbA1c: hemoglobin A1c

MBG: mean glucose

NT-proBNP: N-terminal Pro-B-type Natriuretic Peptide

RMSE: root mean square error

SCORE2-Diabetes: Systematic Coronary Risk Evaluation 2-Diabetes

SMBG: self-monitoring blood glucose

T1D: type 1 diabetes

T2D: type 2 diabetes

TAR: time above range

TBR: time below range

TIR: time in range

Introduction 1

1.1 Diabetes

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.[1,2] The disease is one of the fastest growing global healths challenges.[3] In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.[3] The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).[1] T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β-cells, leading to an absolute insulin deficiency.[4] Approximately 85%-95% of all people with diabetes in developing countries have T2D.[1] It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.[1,5] Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β-cell dysfunction results in inadequate insulin production to meet the body's demands.[1,2,5] T2D is the most which is is Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).[4,6]

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.[4] Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.[4] HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over $\sim$ 120 days lifespan of erythrocytes (red blood cells).[4] However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).[4] With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.[7] Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,[4,8] however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.[4]

1.2 Blood Glucose Monitoring

HbA1c values are a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individuals specific situation HbA1c can vary from e.g., <48mmol/mol to none.[9,10] There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring (CGM) measuring interstitial fluid continuously.[10,11] Both monitoring devices have shown to improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥48 [mmol/mol]	

provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.[10,12,13]

The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.[14] Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, eduction and income.[9,14] CGM devices focuses on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from you device (Figure 1.1). The AGP report shows summary statistics over all your data (usually a two weeks period) including CGM-derived metrics as shown in (Table 1.2).[12] Besides monitoring, data from CGM devices are now used in combination with insulin delivery systems to automate the process of administering insulin.[14] Automated insulin delivery (AID) systems typically use data from CGM devices to determine or adjust insulin delivery. The simplest AIDs are threshold-based low-glucose suspend systems in which insulin delivery is automatically suspended when blood glucose levels drop below a predefined threshold.**[race7?]** More advanced systems utilize additional information such as carbohydrate intake and physical activity to optimize insulin delivery.**[race8?]**

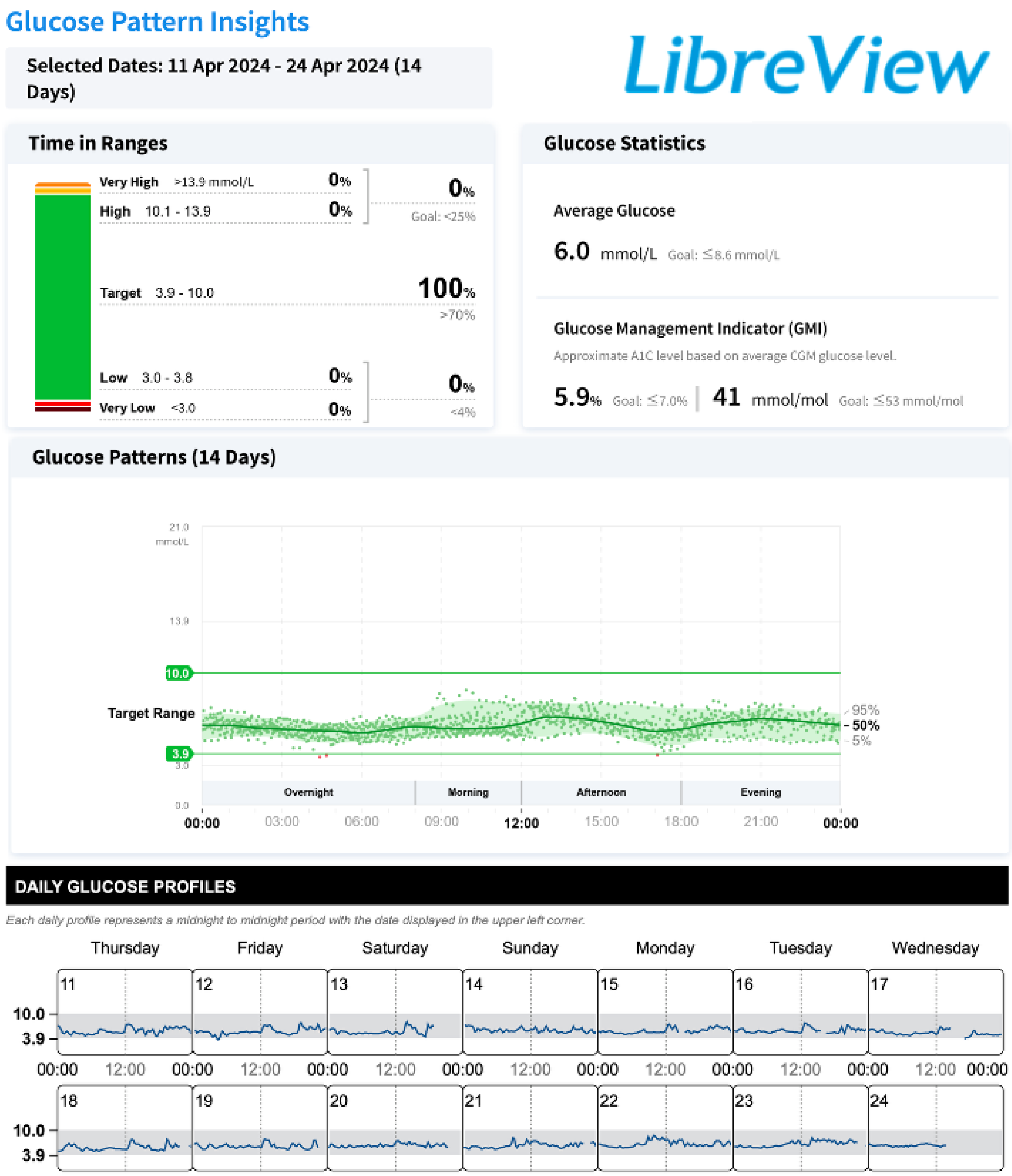

Figure 1.1: Ambulatory glucose profile report from female without diabetes. Data points are from the author herself using an intermittently-scanned (Manuel data upload) FreeStyle LibreLink sensor.

1.3 Glycemic Variability

The challenge in obtaining good glycemic control is to keep blood glucose levels low enough to prevent comorbidities while avoiding the acute symptoms and, for insulin treated people with diabetes, potential mortality of iatrogenic(treatment-induced) hypoglycemia.[15] The variability in the blood glucose fluctuations needs to be reduced in order to lower the average blood glucose levels while still avoiding hypoglycemia.[15] Advances in CGM technology have enabled the quantification of glucose variability, a measure of the extent of fluctuations in blood glucose levels over time,

Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended target
GMI	Glucose Management Indicator		No specific target
TIR	Time in Range	3.9-10.0 mmol/L (70-180 mg/dL)	$> 70\%$
TBR_{54}	Time Below Range	< 3.0 mmol/L (< 54 mg/dL)	$< 1\%$
TBR_{70}	Time Below Range	< 3.9 mmol/L (< 70 mg/dL)	$< 4\%^{*}$
TAR_{180}	Time Above Range	> 10.0 mmol/L (> 180 mg/dL)	$< 25\%^{**}$
TAR_{250}	Time Above Range	> 13.9 mmol/L (> 250 mg/dL)	$< 5\%$
CV	Coefficient of Variation		$\leq 36\%$
	Number of days sensor was worn		≥ 14 days for pattern management
	Active sensor time		70% of 14 days

* This goal include both TBR_{54} and TBR_{70}.
** This goal include both TAR_{180} and TAR_{250}

including both peaks and nadirs.[16] Describing the variability is clinically relevant because two individuals with similar HbA1c levels may exhibit different glycemic variability profiles.[15] Numerous metrics have been developed to characterize different aspects of glucose variability, but no single measure captures all of its dimensions. Consequently, no universal consensus has been reached, although the coefficient of variation (CV) and SD have been incorporated into AGP reports and the American Diabetes Association guidelines (Table 1.2).[15,16]

1.4 Cardiovascular Diseases and Heart Failure

Although advances in diabetes management have reduced the incidence of several diabetes-related complications, cardiovascular disease remains a major cause of morbidity and mortality in people with diabetes.[17–19] Chronic hyperglycemia can lead to long-term complications, although the progression and manifestation of these complications vary between individuals.[4] Identifying individuals at increased risk of atherosclerotic cardiovascular disease or heart failure before clinical disease develops is therefore an important component of diabetes care.[18]

Heart failure, as the condition is at least twice as prevalent among people with diabetes as in the general population.[18,20] It may present as either an acute or a chronic condition which is characterized by a combination of symptoms, including breathlessness, fatigue, and ankle swelling, together with clinical signs such as elevated jugular venous pressure and pulmonary crackles.[21] In individuals with suspected chronic heart failure based on risk factors, symptoms, clinical findings, and supporting investigations such as electrocardiography, measurement of N-terminal Pro-B-type Natriuretic Peptide (NT-proBNP) is recommended as part of the diagnostic work-up.[21] An NT-proBNP concentration of 125 pg/mL indicates the need for further cardiac evaluation, whereas concentrations below this threshold make chronic heart failure unlikely and alternative diagnoses should be considered.[21] Consequently, NT-proBNP represents a clinically relevant biomarker for identifying individuals at increased risk of chronic heart failure.

1.5 Clinical Prediction Models

To facilitate early identification of individuals at increased cardiovascular risk, several clinical prediction models have been developed.[19,22–24] By estimating the probability of a cardiovascular disease (CVD) event within a specified time horizon, these models can increase awareness of cardiovascular risk, encourage adherence to recommended lifestyle changes and medical treatment, and help identify individuals who are most likely to benefit from preventive interventions.[25] Among these, the Systematic Coronary Risk Evaluation 2-Diabetes (SCORE2-Diabetes) model estimates the 10-year risk of CVD in individuals with T2D. It was developed and calibrated using European cohorts, making it particularly relevant for the Danish cohort included in this dissertation.[19] However, the SCORE2-Diabetes model relies solely on traditional clinical variables, including age, sex, smoking status, systolic blood pressure, high-density lipoprotein cholesterol, age at diabetes diagnosis, HbA1c, and Estimated Glomerular Filtration Rate (eGFR), and does not incorporate information on glucose dynamics beyond the long-term glycemic exposure reflected by HbA1c.[19] HbA1c have been independently associated with risk of heart failure,[26] however it does not capture the short-term glucose fluctuations, which may be associated with the progression of vascular complications in diabetes.[27] However association is not prediction and this distinction is important, as association and prediction are fundamentally different objectives, despite the terms sometimes being used interchangeably in the medical literature.[28] Association studies aim to identify relationships between variables and an outcome at group level, whereas prediction models are developed to estimate future outcomes for individual patients.[28]

1.6 Association vs Prediction

Traditional statistical modelling is often used to test hypotheses regarding associations between exposures X and outcomes Y.[29] Regression models applied to observational data are commonly used to evaluate hypotheses derived from prior knowledge, where the choice of exposures is guided by existing theory.[29] These approaches assume that the chosen statistical model adequately represents the underlying data-generating mechanism, allowing conclusions to be drawn from the estimated model parameters.[30] Results from association studies are typically summarized using estimated effect measures, such as regression coefficients, odds ratios, hazard ratios, or risk ratios, together with confidence intervals and significance tests,[28]

In contrast, prediction modelling applies statistical models to predict future outcomes Y based on a set of predictors X.[30] Prediction models may be implemented using a wide range of approaches, including regression models, decision trees, and machine learning algorithms.[29,30] The prediction models are fitted to data (the predictors) and the output $Y\acute{}$ is evaluated based on your outcome Y,[30] and evaluation metrics on prediction studies often include classification metrics such as Area Under the Receiver Operating Characteristic Curve (AUROC), sensitivity, specificity and calibration plots[31] or correlation coefficients, and root mean square error (RMSE), for forecasting models.[32]

1.7 Machine Learning

Associations between CGM-derived metrics and CVD outcomes have been investigated in the literature,[33–35] whereas evidence regarding the predictive value of CGM-derived metrics for cardiovascular risk assessment remains limited.[36,**Hua2012?**,**Koroleva2022?**] With the increased availability of CGM data, machine learning models have become increasingly popular.[37] Machine learning models are often devided into supervised and unsupervised machine learning.[38] Supervised machine learning uses labelled data where each sample (e.g. CGM data from one individual) has a label or an outcome (e.g., elevated NT-proBNP) to predict outcomes for new unforseen data. Unsupervised machine learning on the other hand uses data without labels to explore underlying structures or clusters in data.[38]

especially the neural networks that does not require any manuel processing or feature extraction of the predictors but has automated the process.[37]

Although prediction models differ from association studies in their objective, both have traditionally relied on predefined predictors selected based on prior biological or clinical knowledge. In diabetes research, this includes conventional clinical variables as well as handcrafted CGM-derived metrics such as TIR, MAGE, and coefficient of variation, which were developed to summarize specific aspects of glucose regulation.

Then:

While these metrics are physiologically motivated and clinically interpretable, they represent predefined summaries of the glucose signal and may therefore fail to capture more complex temporal patterns.

Now you've naturally arrived at the limitation.

Then introduce machine learning

Machine learning provides an alternative approach by learning predictive patterns directly from data rather than relying solely on predefined features. Instead of manually engineering summary metrics, machine learning models can learn representations that capture complex characteristics of the underlying glucose signal.

Relevant? Artificial Intelligence: The Future for Diabetes Care: https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S0002934320303399?returnurl=https:%2F%2Flinkinghub.elsevier.com%
Mo2023: says something about glucose variability metrics, what they are and what critiques are there to say about them: Glycemic variability: Measurement, target, impact on complications of diabetes and does it really matter? https://onlinelibrary.wiley.com/doi/10.1111/jdi.14112

- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
  - Logistic regression
  - LSTM and CNNs
  - Foundation models and their advantage
  - The CGMformer is bla bla bla[39]
- Challenges and pitfalls of machine learning in healthcare
  - Overfitting and generalizability
  - Interpretability and explainability

1.8 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care: https://pubmed.ncbi.nlm.nih.gov/16107622/

References

1. Sacks DB, Arnold M, Bakris GL, et al. Guidelines and recommendations for laboratory analysis in the diagnosis and management of diabetes mellitus. Diabetes care 2023;46(10):e151–99.

2. Deshpande AD, Harris-Hayes M, Schootman M. Epidemiology of diabetes and diabetes-related complications. Physical therapy 2008;88(11):1254–64.

3. Magliano DJ, Boyko EJ. IDF diabetes atlas [Internet]. 10th ed. Brussels, Belgium: International Diabetes Federation; 2021. Available from: https://www.ncbi.nlm.nih.gov/books/NBK581930/

4. American Diabetes Association Professional Practice Committee for Diabetes*. 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes—2026. Diabetes Care [Internet] 2025 [cited 2026 July 8];49(Supplement_1):S27–49. Available from: https://doi.org/10.2337/dc26-S002

5. Hall ME, Hall JE. Guyton and hall textbook of medical physiology. Elsevier; 2020.

6. American Diabetes Association Professional Practice Committee. 6. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes—2024. Diabetes Care [Internet] 2023 [cited 2024 May 17];47(Supplement_1):Suppl 1:S111–S125. Available from: https://doi.org/10.2337/dc24-S006

7. Selvin E, Rawlings AM, Bergenstal RM, Coresh J, Brancati FL. No Racial Differences in the Association of Glycated Hemoglobin With Kidney Disease and Cardiovascular Outcomes. Diabetes Care [Internet] 2013 [cited 2026 July 9];36(10):2995–3001. Available from: https://doi.org/10.2337/dc12-2715

8. Kirk JK, D'Agostino RB, Bell RA, et al. Disparities in HbA1c Levels Between African-American and Non-Hispanic White Adults With Diabetes. Diabetes care [Internet] 2006 [cited 2026 July 9];29(9):2130–6. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC3557948/

9. American Diabetes Association Professional Practice Committee for Diabetes*. 6. Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises: Standards of Care in Diabetes—2026. Diabetes Care [Internet] 2025 [cited 2026 July 9];49(Supplement_1):S132–49. Available from: https://doi.org/10.2337/dc26-S006

10. International Diabetes Federation. IDF global clinical practice recommendations for managing type 2 diabetes [Internet]. Brussels, Belgium: International Diabetes Federation; 2025. Available from: https://idf.org/download-global-clinical-practice-recommendations/

11. Buscemi S, Re A, Batsis J A, et al. Glycaemic variability using continuous glucose monitoring and endothelial function in the metabolic syndrome and in Type 2 diabetes. Comment in: Diabet Med 2011 Jan;28(1):125-6; author reply 127-8 PMID: 21166857 [https://wwwncbinlmnihgov/pubmed/21166857] Comment in: Diabet Med 2011 Jan;28(1):126; author reply 127-8 PMID: 21166858 [https://wwwncbinlmnihgov/pubmed/21166858] [Internet] 2010;27(8):872–8. Available from: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=med8&NEWS=N&AN=20653743

12. Danne T, Nimri R, Battelino T, et al. International consensus on use of continuous glucose monitoring. Diabetes care 2017;40(12):1631–40.

13. Freckmann G, Schauer S, Beltzer A, et al. Continuous Glucose Profiles in Healthy People With Fixed Meal Times and Under Everyday Life Conditions. Journal of Diabetes Science and Technology 2024;18(2):407–13.

14. American Diabetes Association Professional Practice Committee for Diabetes*. 7. Diabetes Technology: Standards of Care in Diabetes—2026. Diabetes Care [Internet] 2026 [cited 2026 July 13];49(Supplement_1):S150–65. Available from: https://doi.org/10.2337/dc26-S007

15. Kovatchev B, Cobelli C. Glucose Variability: Timing, Risk Analysis, and Relationship to Hypoglycemia in Diabetes. Diabetes Care [Internet] 2016 [cited 2024 Dec 20];39(4):502–10. Available from: https://doi.org/10.2337/dc15-2035

16. Mo Y, Lu J, Zhou J. Glycemic variability: Measurement, target, impact on complications of diabetes and does it really matter? Journal of Diabetes Investigation [Internet] 2024 [cited 2026 July 13];15(1):5–14. Available from: https://onlinelibrary.wiley.com/doi/10.1111/jdi.14112

17. Rawshani A, Rawshani A, Franz én S, et al. Mortality and Cardiovascular Disease in Type 1 and Type 2 Diabetes. New England Journal of Medicine [Internet] 2017 [cited 2026 July 13];376(15):1407–18. Available from: https://www.nejm.org/doi/full/10.1056/NEJMoa1608664

18. Bajaj M, McCoy RG, Balapattabi K, et al. 10. Cardiovascular disease and risk management: Standards of care in diabetes-2026. Diabetes care 2026;49(Supplement_1):S216–45.

19. Pennells L, Kaptoge S, Østergaard HB, et al. SCORE2-diabetes: 10-year cardiovascular risk estimation in type 2 diabetes in europe. European heart journal 2023;44(28):2544–56.

20. Hoek AG, Dal Canto E, Wenker E, et al. Epidemiology of heart failure in diabetes: A disease in disguise. Diabetologia [Internet] 2024 [cited 2026 July 13];67(4):574–601. Available from: https://doi.org/10.1007/s00125-023-06068-2

21. McDonagh TA, Metra M, Adamo M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) With the special contribution of the Heart Failure Association (HFA) of the ESC. European Heart Journal [Internet] 2021 [cited 2026 July 13];42(36):3599–726. Available from: https://doi.org/10.1093/eurheartj/ehab368

22. Hemann BA, Bimson WF, Taylor AJ. The framingham risk score: An appraisal of its benefits and limitations. American Heart Hospital Journal [Internet] 2007;5(2):91–6. Available from: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1541-9215.2007.06350.x

23. Hippisley-Cox J, Coupland C, Vinogradova Y, Robson J, May M, Brindle P. Derivation and validation of QRISK, a new cardiovascular disease risk score for the United Kingdom: Prospective open cohort study. BMJ : British Medical Journal [Internet] 2007 [cited 2026 July 13];335(7611):136. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC1925200/

24. Segar MW, Vaduganathan M, Patel KV, et al. Machine Learning to Predict the Risk of Incident Heart Failure Hospitalization Among Patients With Diabetes: The WATCH-DM Risk Score. Diabetes Care [Internet] 2019 [cited 2026 July 13];42(12):2298–306. Available from: https://doi.org/10.2337/dc19-0587

25. Lloyd-Jones DM. Cardiovascular risk prediction. Circulation [Internet] 2010;121(15):1768–77. Available from: https://www.ahajournals.org/doi/abs/10.1161/CIRCULATIONAHA.109.849166

26. Hsiao F-C, Lin C-P, Hsu T-J, et al. Glycohemoglobin variations and long-term risk of incident heart failure in patients with type 2 diabetes. Mayo Clinic proceedings 2025;100(8):1332–44.

27. Belli M, Bellia A, Sergi D, Barone L, Lauro D, Barillà F. Glucose variability: A new risk factor for cardiovascular disease. Acta diabetologica 2023;60(10):1291–9.

28. Varga TV, Niss K, Estampador AC, Collin CB, Moseley PL. Association is not prediction: A landscape of confused reporting in diabetes – A systematic review. Diabetes Research and Clinical Practice [Internet] 2020 [cited 2025 Aug 6];170:108497. Available from: https://www.sciencedirect.com/science/article/pii/S0168822720307518

29. Shmueli G. To Explain or to Predict? Statistical Science [Internet] 2010 [cited 2026 July 14];25(3):289–310. Available from: https://www.jstor.org/stable/41058949

30. Breiman L. Statistical Modeling: The Two Cultures (with comments and a rejoinder by the author). Statistical Science [Internet] 2001 [cited 2026 July 14];16(3):199–231. Available from: https://projecteuclid.org/journals/statistical-science/volume-16/issue-3/Statistical-Modeling--The-Two-Cultures-with-comments-and-a/10.1214/ss/1009213726.full

31. Van Calster B, Collins GS, Vickers AJ, et al. Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: Overview and guidance. The Lancet Digital Health [Internet] 2025;7(12):100916. Available from: https://www.sciencedirect.com/science/article/pii/S2589750025000986

32. Doorn WPTM van, Foreman YD, Schaper NC, et al. Machine learning-based glucose prediction with use of continuous glucose and physical activity monitoring data: The maastricht study. PloS one 2021;16(6):e0253125–.

33. Yapanis M, James S, Craig ME, O'Neal D, Ekinci EI. Complications of diabetes and metrics of glycemic management derived from continuous glucose monitoring. The journal of clinical endocrinology and metabolism 2022;107(6):e2221–36.

34. Thomsen HB, Lebiecka-Johansen B, Nørgaard O, et al. Continuous glucose monitoring–derived metrics and cardiovascular risk among people with diabetes: Systematic scoping review. JMIR Diabetes [Internet] 2026;11:e89374. Available from: https://doi.org/10.2196/89374

35. Zhang L, Sun X-X, Tian Q. Research progress on the association between glycemic variability index derived from CGM and cardiovascular disease complications. Acta diabetologica 2024;61(6):679–92.

36. Su G, Mi S, Tao H, et al. Association of glycemic variability and the presence and severity of coronary artery disease in patients with type 2 diabetes. Cardiovascular diabetology 2011;10(1):19–9.

37. Jaloli M, Cescon M. Long-Term Prediction of Blood Glucose Levels in Type 1 Diabetes Using a CNN-LSTM-Based Deep Neural Network. Journal of Diabetes Science and Technology [Internet] 2023 [cited 2025 Jan 28];17(6):1590–601. Available from: http://journals.sagepub.com/doi/10.1177/19322968221092785

38. Badillo S, Banfai B, Birzele F, et al. An Introduction to Machine Learning. Clinical Pharmacology & Therapeutics [Internet] 2020 [cited 2026 July 14];107(4):871–85. Available from: https://ascpt.onlinelibrary.wiley.com/doi/10.1002/cpt.1796

39. Lu Y, Liu D, Liang Z, et al. A pretrained transformer model for decoding individual glucose dynamics from continuous glucose monitoring data. National science review 2025;12(5):nwaf039–.

Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose Self-Monitoring/	10241
2	Continuous Glucose Monitoring/	34
3	(((blood-glucose or bloodglucose or glucose) adj2 (monitor$ or selfmonitor))orcgmorbgmoriscgmorrtcgmor(timeadj3range)).ti, ab, kf, ku or bloodsugar$ or glucose$ or glyc?emic$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent$ or scan$ or variabilit).ti, ab, kf, kw.|8669||5|or/1–4|38366||6|expCardiovascularDiseases/|2760764||7|(cardiovasc or cvd or cardiometabol$ or cardio-metabol$ or heart$ or macrovasc$ or vasc).ti, ab, kw, kf.|2182473||8|or/6–7|3971342||9|expDiabetesMellitus/|518864||10|(diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13	2300
	Updated search: 11 March 2025	
15	limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")	310
16	or/14-15	2610

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

<table>
<tr><th>#</th><th>Searches</th><th>Results</th></tr>
<tr><td>1</td><td>blood glucose monitoring/</td><td>37574</td></tr>
<tr><td>2</td><td>exp continuous glucose monitoring system/</td><td>5648</td></tr>
<tr><td>3</td><td>(((blood-glucose or bloodglucose or glucose) adj2 (monitor$ or selfmonitor))orcgmorbgmoriscgmorrtcgmor(timeadj3range)).ti, ab, kf, kw.|4850 or bloodsugar$ or glucose$ or glyc?emic$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent$ or scan$ or variabilit).ti, ab, kf, kw.|16430||5|or/1−4|81510||6|expcardiovasculardisease/|5122163||7|(cardiovasc or cvd or cardiometabol$ or cardio-metabol$ or heart$ or macrovasc$ or vasc).ti, ab, kw, kf.|3044925||8|or/6−7|6303633||9|expdiabetesmellitus/|1239184||10|(diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.</td><td>1237019</td></tr>
<tr><td>11</td><td>or/9-10</td><td>1501812</td></tr>
<tr><td>12</td><td>and/5,8,11</td><td>11217</td></tr>
<tr><td>13</td><td>(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset$1).ti. and animal experiment/</td><td>1237966</td></tr>
<tr><td>14</td><td>animal experiment/ not (human experiment/ or human/)</td><td>2601109</td></tr>
<tr><td>15</td><td>or/13-14</td><td>2672240</td></tr>
<tr><td>16</td><td>12 not 15</td><td>10861</td></tr>
<tr><td>17</td><td>limit 16 to exclude medline journals
Updated search: 11 March 2025</td><td>2176</td></tr>
<tr><td>18</td><td>limit 17 to (dc=“20231201-20261231” or dd=“20231201-20261231” or rd=“20231201-20261231”)</td><td>467</td></tr>
<tr><td>19</td><td>or/17-18</td><td>2643</td></tr>
</table>

B Study I: Included and excluded reports

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results D

- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results

E Study II: Prediction stability

Prediction stability plots (**?@fig-pred_stability**) revealed more stability the fewer predictors.

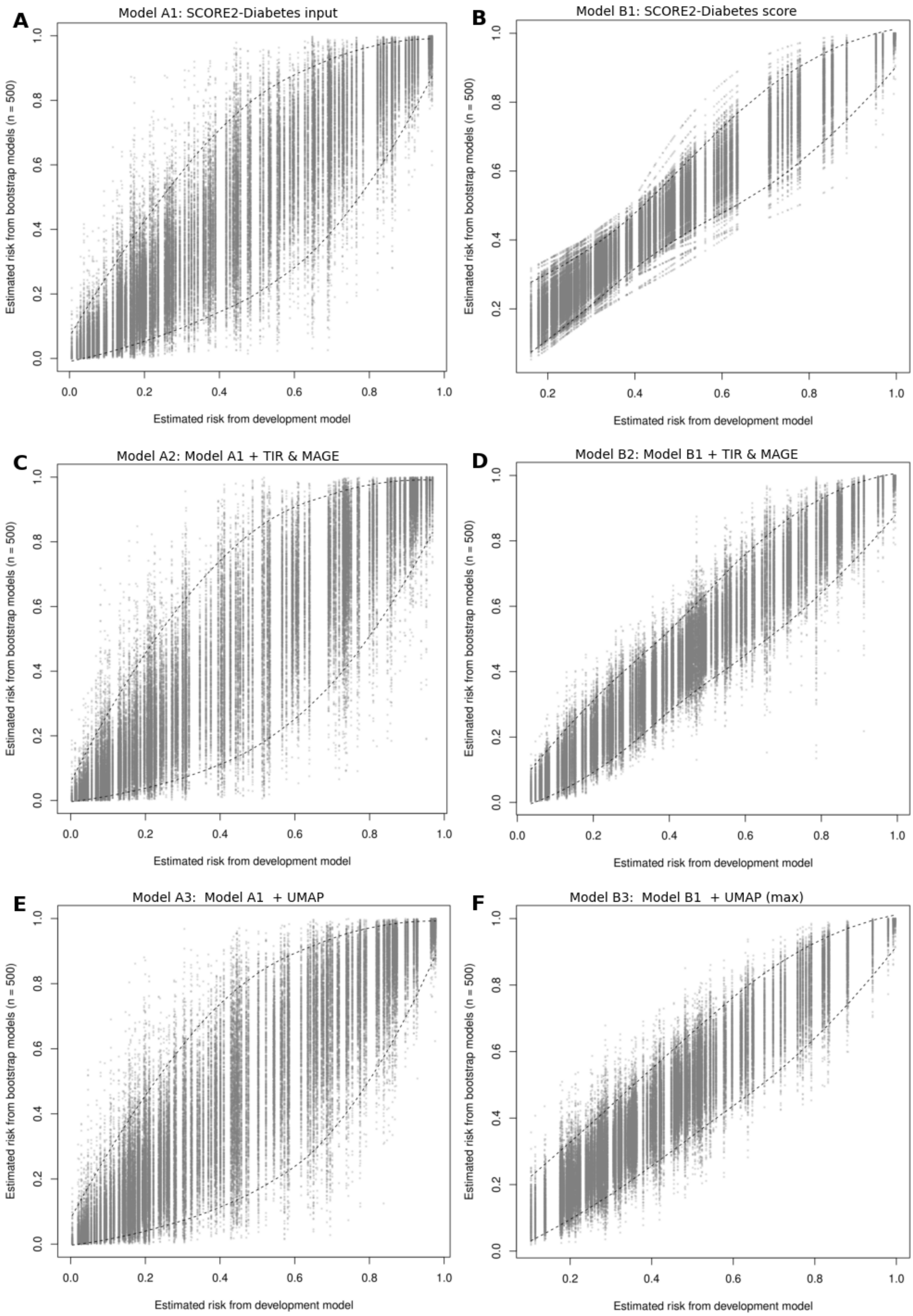

Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

☐ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
		Extreme	0%	0%	0%	0%	0%	0%
	Base -Generalized	None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III

Addenda and errata H

This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation.
- Errata: corrections to errors discovered since submission of the dissertation.

H.1 Addenda

H.2 Errata